

6.8 Quiz: Polynomials Challenge (no calculator)

1. [Maximum mark: 7]

Consider the function $f(x) = -2(x + 3)(x - 5)$, for $x \in \mathbb{R}$.

(a) For the graph of f

(i) write down the x -intercepts;

[2]

(ii) find the coordinates of the vertex.

[3]

The function f can be written in the form $f(x) = -2(x - h)^2 + k$.

(b) Write down the value of h and the value of k .

[2]

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2. [Maximum mark: 7]

The line $y = mx$ passes through the origin and is tangent to the parabola $y = x^2 + 16$.

(a) Show that at any point of intersection, $x^2 - mx + 16 = 0$. [1]

(b) Find the discriminant of $x^2 - mx + 16 = 0$ in terms of m . [1]

(c) Find the values of m for which the line is tangent to the parabola. [2]

(d) For each value of m , find the coordinates of the point of tangency. [3]

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3. [Maximum mark: 7]

Consider the function $f(x) = 3x^2 + 6x - 2$.

(a) Write $f(x)$ in the form $a(x - h)^2 + k$. [3]

(b) Write down the coordinates of the vertex. [1]

(c) Find the discriminant of $f(x) = 0$. [1]

(d) State the number of x -intercepts of the graph and justify your answer. [2]

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4. [Maximum mark: 8]

Let $p(x) = 2x^3 - 3x^2 - 11x + 6$.

(a) Show that $(x - 3)$ is a factor of $p(x)$. [2]

(b) Hence express $p(x)$ in the form $(x - 3)(ax^2 + bx + c)$. [2]

(c) Factorise $p(x)$ completely into three linear factors. [2]

(d) Write down all solutions to $p(x) = 0$. [1]

(e) Write down the y -intercept of the graph $y = p(x)$. [1]